

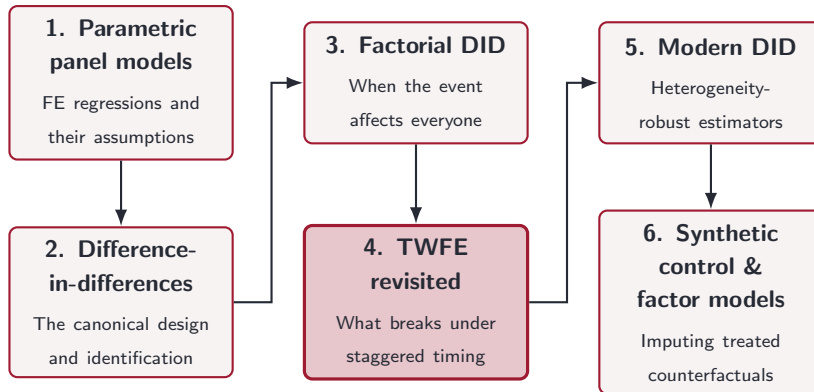
# Panel Methods (4): Two-Way Fixed Effects Revisited

Lectures on Causal Panel Analysis

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- Lecture 1: parametric panel models, with FE regressions as the workhorse
- Lecture 2: DID as a research design, mostly with two groups and fixed treatment timing
- Lecture 3: factorial DID, extending the design to events that hit every unit
- In the two-period case, the TWFE estimate *is* the DID estimate, so the two traditions seem interchangeable
- Today: what happens when we run TWFE regressions on **many periods**, **staggered timing**, and **heterogeneous effects**, which is what applied work actually looks like

- What assumptions do TWFE models rely upon?
- Three problems
  - Implications for treatment assignment (design)
  - Parallel trends violations
  - The weighting problem

## Question

When we estimate  $Y_{it} = \tau D_{it} + \alpha_i + \xi_t + \varepsilon_{it}$  on a staggered panel, which comparisons is  $\hat{\tau}$  actually making?

- Two groups ( $\mathcal{T}, \mathcal{C}$ ), two periods ( $t$  and  $t'$ ), fixed treatment timing
- Functional form:

$$Y_{it} = \tau_{it}D_{it} + \alpha_i + \xi_t + \varepsilon_{it} \quad \text{or} \quad \begin{cases} Y_{it}(0) = \alpha_i + \xi_t + \varepsilon_{it} \\ Y_{it}(1) = Y_{it}(0) + \tau_{it} \end{cases}$$

where  $\tau_{it}$  is the treatment effect for unit  $i$  at time  $t$

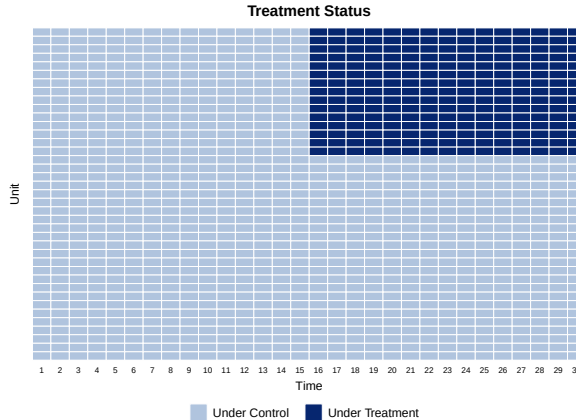
- Parallel trends:  $\mathbb{E}[Y_{it'}(0) - Y_{it}(0) \mid i \in \mathcal{T}] = \mathbb{E}[Y_{jt'}(0) - Y_{jt}(0) \mid j \in \mathcal{C}]$
- Or equivalently,  $\mathbb{E}[\varepsilon_{it'} - \varepsilon_{it} \mid i \in \mathcal{T}] = \mathbb{E}[\varepsilon_{jt'} - \varepsilon_{jt} \mid j \in \mathcal{C}]$
- $ATT = \mathbb{E}[\tau_{it} \mid D_{it} = 1]$  is nonparametrically identified with two periods (or two treatment histories)

- The treated-group untreated potential outcome after treatment is the one cell we never observe:

|                              | Pre                      | Post                      |
|------------------------------|--------------------------|---------------------------|
| Treated ( $\mathcal{T}$ )    | $Y_{\mathcal{T}, pre}^0$ | ??                        |
| Comparison ( $\mathcal{C}$ ) | $Y_{\mathcal{C}, pre}^0$ | $Y_{\mathcal{C}, post}^0$ |

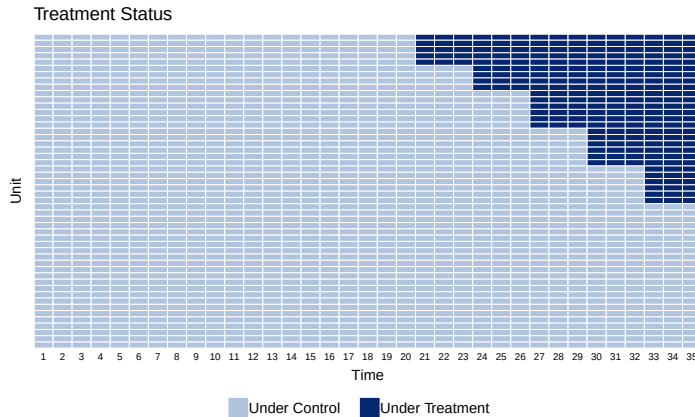
- Parallel trends fills in the missing cell: take the treated pre-period level and add the comparison group's change
- Every estimator in these lectures is a different rule for filling in that cell

## Beyond Two Periods: Block Assignment



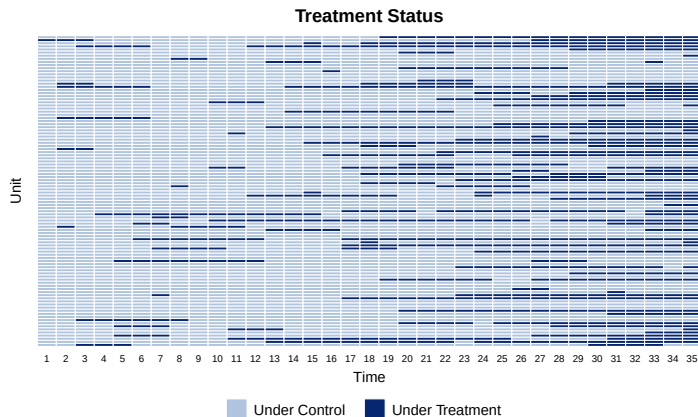
- One adoption date, many periods: still essentially a  $2 \times 2$  comparison

## Beyond Two Periods: Staggered Adoption



- Different units adopt at different times: the classic staggered setting

## Beyond Two Periods: The General Case (with Treatment Reversal)



- Treatment can also switch off; the majority of political science applications look like this (we return to it in Lecture 5)

$$Y_{it} = \tau D_{it} + \mathbf{X}_{it}'\boldsymbol{\beta} + \alpha_i + \xi_t + \varepsilon_{it}, \quad D_{it} \in \{0, 1\}$$

- **Functional form**

- Additive fixed effects
- **Constant** and contemporaneous treatment effect
- Linearity in covariates

- Note the difference from the DID setup: DID allowed  $\tau_{it}$  to vary; the TWFE regression assumes a single  $\tau$

⇒ When effects are actually heterogeneous, this assumption fails and induces the **(negative) weighting** problem

- **Strict exogeneity:**

$$\varepsilon_{it} \perp\!\!\!\perp D_{js}, \mathbf{X}_{js}, \alpha_j, \xi_s \quad \forall i, j, t, s$$

- Which implies

$$\{Y_{it}(0), Y_{it}(1)\} \perp\!\!\!\perp D_{js} \mid \mathbf{X}, \alpha, \xi \quad \forall i, j, t, s$$

- With only two groups, this delivers parallel trends:

$$\mathbb{E}[Y_{it}(0) - Y_{it'}(0) \mid \mathbf{X}] = \mathbb{E}[Y_{jt}(0) - Y_{jt'}(0) \mid \mathbf{X}], \quad i \in \mathcal{T}, j \in \mathcal{C}, \forall t, t'$$

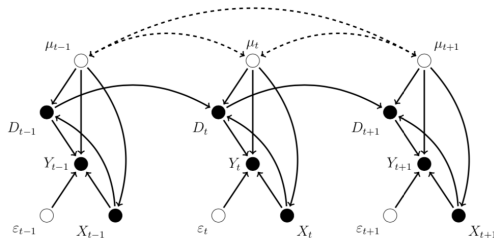
$$Y_{it} = \tau D_{it} + \mathbf{X}_{it}'\boldsymbol{\beta} + \alpha_i + \xi_t + \varepsilon_{it}$$

- Assumptions
  - Functional form
  - Strict exogeneity
- Challenges
  - Strict exogeneity means a lot more than you think
  - Prevalent parallel trends failures
  - Treatment effect heterogeneity leads to bias

We take the three in turn: the design problem, then parallel trends, then weighting.

# Strict Exogeneity Means More Than You Think

Reinterpreting strict exogeneity (Imai and Kim 2019):



- 1 No unobserved **time-varying** confounder exists
- 2 No lagged dependent variable  $\rightarrow$  adding an LDV is fine if  $T$  is large; not adding it is also fine
- 3 No “carryover effect”  $\rightarrow$  adding lagged treatment terms is fine
- 4 Past outcomes do not directly affect current treatment (no “feedback”)

Each of these is a substantive claim about the world, not a regression option.

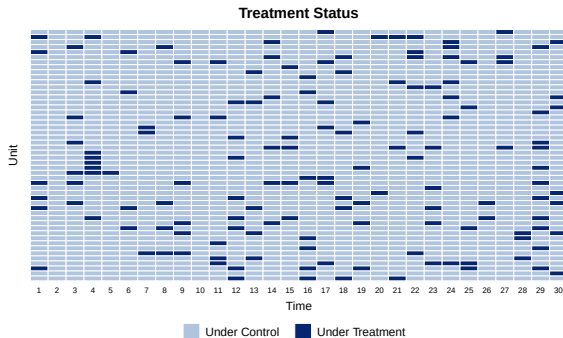
# Hypothetical Experiment?

DGPs consistent with strict exogeneity:

$$\alpha_i, X_i \rightarrow D_i \rightarrow Y_i$$

Treatment status is assigned randomly or in one shot, not sequentially!

Examples: **random assignment** within units



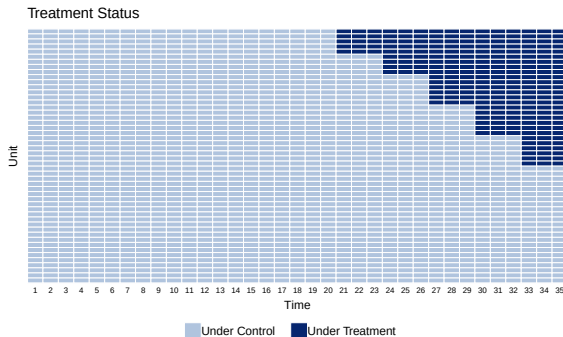
# Hypothetical Experiment?

DGPs consistent with strict exogeneity:

$$\alpha_i, \mathbf{X}_i \rightarrow A_i \rightarrow D_i \rightarrow Y_i$$

Treatment status is assigned randomly or in one shot, not sequentially!

Examples: **staggered adoption**, with  $A_i$  the adoption time (Athey and Imbens 2022)



## The Event-Study (Dynamic TWFE) Specification

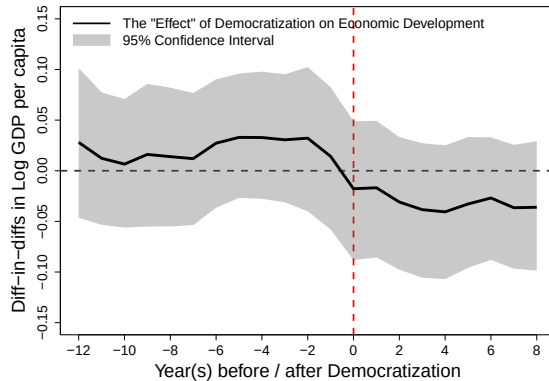
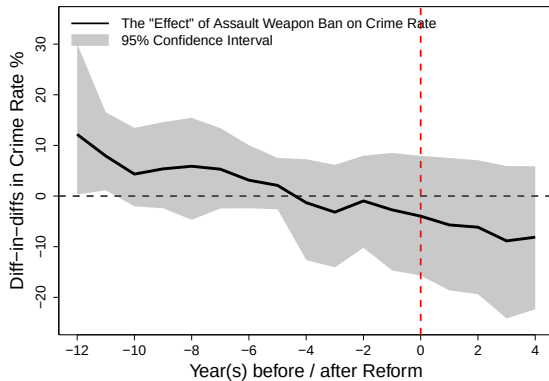
- To probe parallel trends and trace dynamics, applied work estimates

$$Y_{it} = \sum_{s \neq -1} \mu_s \mathbf{1}\{t - T_i = s\} + \alpha_i + \xi_t + \varepsilon_{it}$$

where  $T_i$  is unit  $i$ 's adoption date and  $s$  counts periods relative to adoption

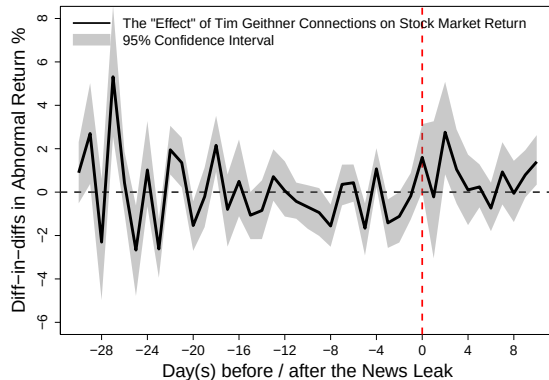
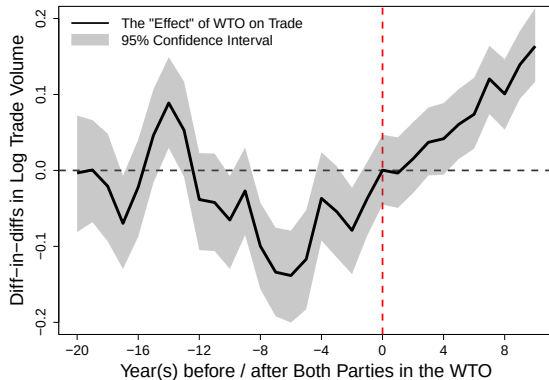
- The period just before adoption ( $s = -1$ ) is the omitted reference category
- **Leads** ( $s < -1$ ) probe pre-treatment differences; **lags** ( $s \geq 0$ ) trace the dynamic effects
- Endpoints are usually binned (e.g.  $s \leq -8$ ,  $s \geq 12$ ) so each coefficient rests on enough observations

## Failure of Parallel Trends Is Common (1)



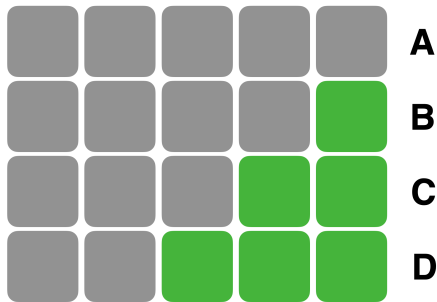
- Pre-treatment differences trend before the treatment ever arrives

## Failure of Parallel Trends Is Common (2)



- Lesson: always plot the pre-treatment differences before interpreting the post-treatment ones

## What's the Weighting Problem?



- Question: can TWFE recover (some **non-negatively weighted**) ATT when treatment effects are heterogeneous?
- Probably not! e.g. Goodman-Bacon (2021), de Chaisemartin and D'Haultfoeulle (2020), Callaway and Sant'Anna (2021), Borusyak et al. (2024), Strezhnev (2018), Imai and Kim (2021)
- Early adopters (e.g. D) serve as comparisons for late adopters (e.g. B)

⇒ Some **treated** observations are used as comparisons

- Under staggered adoption, the TWFE estimator

$$Y_{it} = \beta^{TWFE} D_{it} + \alpha_i + \xi_t + \varepsilon_{it}$$

is a **weighted average of all possible  $2 \times 2$  DID estimators** that compare timing groups to each other

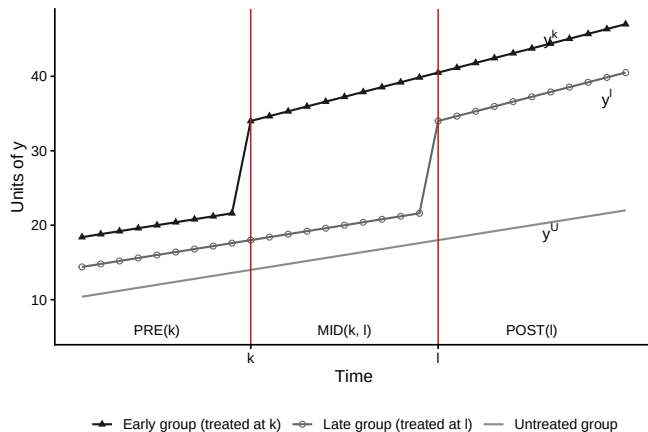
- The weights are proportional to timing-group sizes and the variance of the treatment dummy in each pair, which is highest for units treated in the middle of the panel
- Source of biases:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}^{TWFE} = VWATT + VWCT - \Delta ATT$$

- $VWATT$ : variance-weighted ATT
- $VWCT$ : variance-weighted common trends
- $\Delta ATT$ : change in treatment effects over time

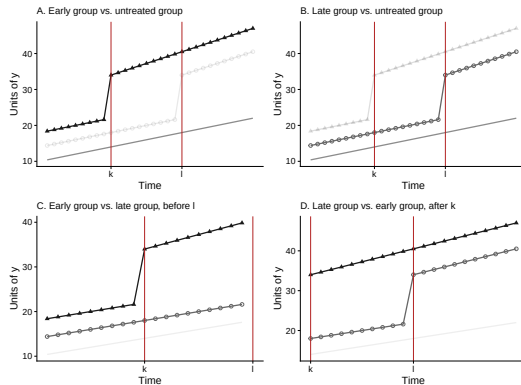
# The Three-Group Case

- An early group  $k$ , treated at  $t_i = k$
- A late group  $\ell$ , treated at  $t_i = \ell > k$
- An untreated group  $U$ ,  $t_i = \infty$



Source: Redrawn after Goodman-Bacon (2021), Figure 1 (goodman\_bacon\_2026.R).

## Four Simple ( $2 \times 2$ ) DID Estimates

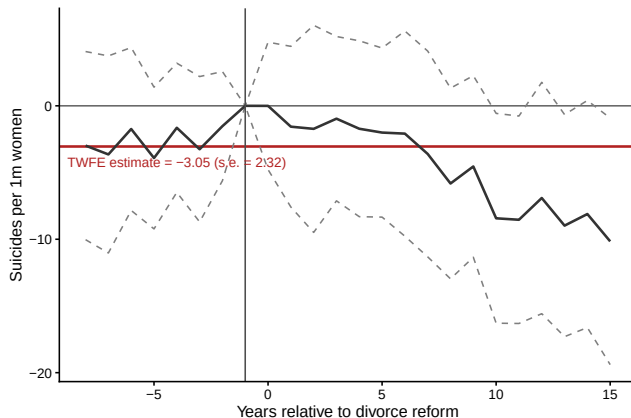


- Panel D is the troublemaker: the **already-treated** early group serves as the comparison, and its evolving treatment effect enters with a **negative sign**

Source: Goodman-Bacon (2021).

## Example: No-Fault Divorce Reforms and Female Suicide

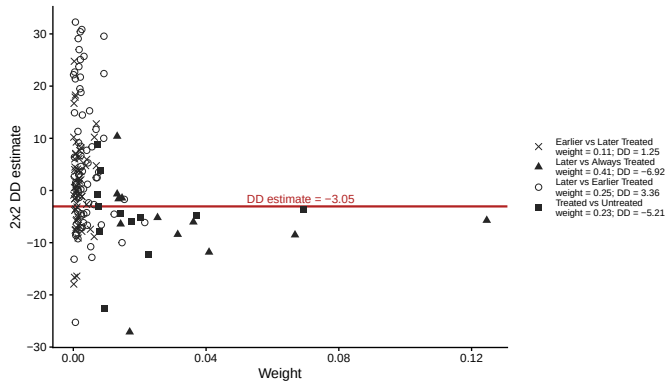
- 37 states adopted unilateral divorce between 1969 and 1985 (9 had reformed before 1964, 5 never did); event-study and TWFE estimates:



Source: Own replication of the divorce example in Goodman-Bacon (2021); Stevenson and Wolfers (2006) data from the `bacondecomp` package

## Decomposing the Divorce Estimate

- Plot each  $2 \times 2$  DID against its weight; average effect and total weight by comparison type:



- The TWFE estimate,  $-3.05$  (Goodman-Bacon reports  $-3.08$ ), is the average of the y-axis values weighted by the x-axis values

Source: Own replication with the `bacondecomp` package (goodman-bacon 2026.R).

- A complementary result: under heterogeneous effects, the TWFE coefficient satisfies

$$\hat{\beta}^{TWFE} \xrightarrow{p} \mathbb{E} \left[ \sum_{(i,t): D_{it}=1} w_{it} \tau_{it} \right]$$

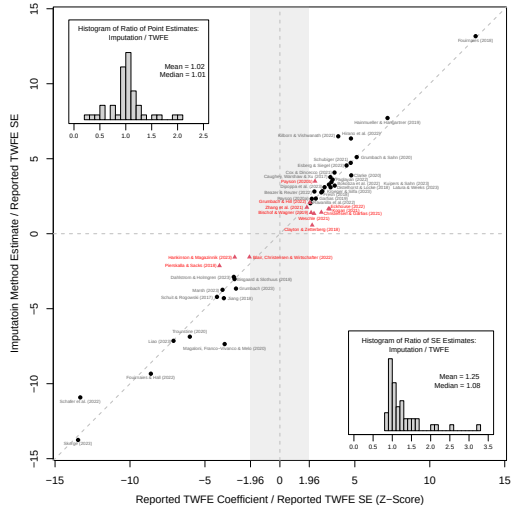
where the weights  $w_{it}$  sum to one but **some can be negative**

- Negative weights attach to observations treated for a long time in periods when many units are treated
- With negative weights,  $\hat{\beta}^{TWFE}$  can be negative even if every  $\tau_{it} > 0$ : a **sign reversal**
- Diagnostic: the `twowayfweights` package reports the share and sum of negative weights for your design

- The same logic contaminates the dynamic specification: each lead or lag coefficient  $\hat{\mu}_s$  is a weighted average that picks up effects from **other relative periods** and **other cohorts**
- In particular, lead coefficients can appear non-zero even when parallel trends holds, and can appear zero when it fails
- So under staggered adoption with HTE, the standard event-study plot is not a reliable pre-trend diagnostic
- Fix: estimate cohort-specific effects and aggregate them explicitly, e.g. the interaction-weighted estimator (Lecture 5)

# How Bad Is the Weighting Problem in Practice? (Chiu et al. 2026)

- Not too bad!
- Reanalyzing published studies from top political science journals, we find:
  - There is almost no sign flipping
  - No systematic overestimation or underestimation of the treatment effects
  - Some marginally significant results become insignificant
  - However, parallel trends violations remain a major concern
- Lecture 5 returns to this study in detail



- TWFE on staggered data mixes clean and contaminated comparisons; with heterogeneous effects, the estimand is no longer any ATT we care about
  - In practice, the first-order threat is usually **parallel trends failure**, not the weighting problem itself
  - Both problems have the same root: the regression does not distinguish *which* observations serve as comparisons for *which* treated observations
  - The fix: make the comparisons explicit, which is what the modern estimators do
- ⇒ The road ahead: Lecture 5 repairs the **weighting problem**; Lecture 6 introduces low-rank methods that can address **some** parallel trends violations.

- **Condition on pre-treatment covariates:** parallel trends holds within strata of  $X$ 
  - Semiparametric DID, e.g. Abadie (2005); Callaway and Sant'Anna (2021) → Lecture 5
- **Condition on pre-treatment outcomes:** build the comparison from units whose pre-trends match
  - Synthetic control, e.g. Abadie et al. (2010, 2015) → Lecture 6
- **Model the confounding:** impute  $Y_{it}(0)$  from a low-rank (latent factor) model fit on untreated data
  - Gobillon and Magnac (2016); Xu (2017); Athey et al. (2021) → Lecture 6
- **Combine weighting and modeling:** doubly robust hybrids
  - Augmented synthetic control (Ben-Michael et al. 2021); synthetic DID (Arkhangelsky et al. 2021) → Lecture 6

- TWFE models require stronger assumptions than we normally admit; both treatment effect heterogeneity and parallel trends failures can lead to bias
- Under staggered adoption, TWFE averages all  $2 \times 2$  comparisons, including ones that use already-treated units as comparisons
- In reanalyses of published work, the weighting bias is usually modest; parallel trends violations are the bigger threat
- Current solutions: matching/reweighting, imputation, and their combinations
- Whichever method you use, run the diagnostics: plot the pre-trend and run placebo tests. They can reject parallel trends, but they cannot prove it.

- `fixest`: fast fixed effects regressions
- `bacondecomp`: Goodman-Bacon decomposition
- `did`: Callaway and Sant'Anna's estimator
- `PanelMatch`: panel matching
- `fect`: imputation estimators (FEct, IFect, MC) with diagnostic tests
- `gsynth`: interactive fixed effects with non-reversible treatments
- `panelView`: panel data visualization

Lectures 5 and 6 demonstrate several of these in worked examples.

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